Expo Poster Panels Reference

Last Updated: 2026-03-05
Purpose: Content guide for two tri-fold poster boards (6 panels total)

TRI-FOLD 1: THE SYSTEM

Left Panel - The Problem

Title: "Why Healthcare IoT Security Matters"

Content:

- **The Crisis:** 88% healthcare data breaches from IoT, $10.93M average cost
- **Real Examples:** 2017 FDA pacemaker recall, 2020 UVM ransomware
- **Our Mission:** Build functional hospital network → demonstrate vulnerabilities → prove defense strategies

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Center Panel - Network Architecture

Title: "A Real Hospital Network in Miniature"

Content:

VLAN 30 (Secure Hospital - 10.30.0.0/24): - 5× Raspberry Pi heart rate monitors → MQTT → Node-RED → PostgreSQL (397M+ records) - Keycloak SSO (4 apps), Wazuh SIEM (14 agents), Suricata IDS - HAPI FHIR server with OAuth2 protection

VLAN 2 (Attack Lab - 192.168.50.0/24): - Pi #7 (vulnerable balloon demo, Switch Port 15) - Port mirroring: Port 6 mirrors Port 15 for IDS monitoring - Admin laptop (10.30.0.123) for attack simulation

Firewall Segmentation: VLAN30 → VLAN2 (admin only), VLAN2 VLAN30 (blocked)

Right Panel - Data Pipeline & Security

Title: "From Heartbeat to FHIR Record"

Content:

Pipeline: SenseHat → MQTT → Node-RED → PostgreSQL + Mirth → FHIR Server → Grafana

Security Stack: - Keycloak (SSO, 80% coverage) - Vault (3-of-5 unseal, secrets management) - Wazuh (centralized logging, anomaly detection) - Suricata (IDS on bare-metal, port mirroring)

Scale: 100 readings/sec, 397M+ records, 7-day retention + 87GB archive

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TRI-FOLD 2: THE DEMO

Left Panel - The Vulnerable System

Title: "How Sensor Spoofing Works"

Content:

Hardware Chain: - DATAQ 100 PSI sensor → ADS1115 ADC → Raspberry Pi #7 → Flask API (port 5000) - Sequent relay HATs → pump (23A) + 3× solenoid valves - 1L plastic bottle pressure vessel

The Vulnerability: - Unauthenticated `/sensor` POST endpoint - No input validation or rate limiting - Trusts network pressure readings blindly

The Attack: Laptop sends fake "0.0 PSI" at 10 Hz → system believes bottle is empty → keeps pump running → real pressure exceeds safe limits → **BURST**

Center Panel - Live Demo

Title: "Vulnerable vs. Secure: Side-by-Side"

Content:

LEFT - Vulnerable System (Pi #7): - Network sensor readings (HTTP API) - No authentication - No input validation - Attack sends 0.0 PSI → pump runs continuously → pressure climbs dangerously

RIGHT - Secure System (Pi #6): - Direct sensor reading (I2C, no network exposure) - Input validation (0.5-4.5V range check) - Rate-of-change detection (5.0 PSI/sec threshold) - Dual-sensor cross-validation - Wazuh SIEM logging - Same attack → anomaly detected → pump disabled → pressure controlled

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Right Panel - Defense Strategies

Title: "Lessons Learned: Hardening IoT Systems"

Content:

What Worked: 1. **Input Validation** - range checks, sanity checks 2. **Anomaly Detection** - rate-of-change thresholds 3. **Sensor Redundancy** - cross-validation between dual sensors 4. **Network Segmentation** - firewall isolation prevents lateral movement 5. **Centralized Logging** - Wazuh SIEM for forensics

Key Takeaways: - Never trust network data without validation - Physical sensors should connect directly when possible - Defense-in-depth: no single point of failure - Real-time monitoring catches attacks before damage

Real-World Impact: These techniques apply to pacemakers, insulin pumps, ventilators, and all critical medical IoT devices.